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Project Write Up

- What are the algorithms that you have used for Matching?
    - During this process I used a rule-based entity matching approach. I first applied blocking using the release_year attribute. That only allowed tuples with the same year were compared. The fuzzy string similarity (RapidFuzz token_sort_ratio) was used to compare textual attributes. If the similarity score exceeded the threshold, the tuple pair was considered a match and added to Table C.
- What are the algorithms that you have used for Matching?
    - I have several data quality issues. These were observed in both tables. Table A contained tuples that did not represent movies, such as person pages or list entries. In addition, some attributes between Table A and Table B were not perfectly aligned, and text values contained inconsistencies such as formatting difference or missing values. Because of these issues, fuzzy matching was used instead of exact matching.
- Report the size of table A, the size of table B, the total number of tuple pairs in the Catersian product of A and B, and the total number of tuple pairs in the table C.
    - Size of Table A: 1000
    - Size of Table B: 1773
    - Total tuple pairs in the cartesian product, 1,773,000
    - Total tuple pairs in Table C (matches found): 197
- Did you have to do any cleaning or additional information extraction on tables A and B?
    - Yes. Table A was filtered to keep only rows that appeared to represent movies by requiring non-null values in the runtime_minutes attribute. Blocking was also applied using release_year to reduce the number of comparisons. In addition, text values were converted to strings before computing similarity scores to handle formatting inconsistencies.